

JUNTANG WANG

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EDUCATION

Harvard University

M.S. in Data Science

2026 – 2028

Cambridge, MA

Duke University & Duke Kunshan University (Dual Degree)

B.S. Interdisciplinary Studies; B.S. Applied Math & Comp. Sci. (CS Track)

Durham, NC; Kunshan, China

2022 – 2026

- Graduation with Distinction (Top 5%) and Latin Honors *cum laude*; GPA: 3.8/4.0 (Duke: 4.0/4.0; Major: 4.0/4.0).
- Thesis: “Manifold-Informed NNs”. A+: Deep Learning, Machine Learning, Graph-Matrix-Network Analysis, Databases.

PUBLICATIONS & OPEN-SOURCE SOFTWARE

[†]Equal contribution. 7 additional first/co-first manuscripts under review (5× KDD, 2× NeurIPS).

Juntang Wang[†], Yihan Wang[†], Hao Wu[†], Jiayu Gao, Shixin Xu, and Dongmian Zou. (2026). “StageGuard: Physiologically Constrained Sleep Staging.” *KDD 2026 AI for Science Track (Inaugural)*. [arXiv:2607.23284](https://arxiv.org/abs/2607.23284)

Juntang Wang[†], Hao Wu[†], Yihan Wang[†], Dongmian Zou, and Shixin Xu. (2026). “Mixing Configurations for Downstream Prediction.” *ICML 2026*. icml.cc/virtual/2026/poster/63010

Juntang Wang[†], Yihan Wang[†], Hao Wu[†], Dongmian Zou, and Shixin Xu. (2025). “Brain-Inspired Perspective on Configurations: Unsupervised Similarity and Early Cognition.” *BICS (Oral)*. [arXiv:2510.19229](https://arxiv.org/abs/2510.19229)

Shu Kit Eric Tam, Juntang Wang, Jianke Cen, Aleksandra Stryjska, Pascal Grange, and Sze Chai Kwok. (2025). “Can the mammalian circadian system adapt to the Martian photoperiod?” *18th Annual Meeting of the Chinese Neuroscience Society*.

Yihan Wang[†], Juntang Wang[†], Xinze Xu, Yihen Han, Qinyi Chen, Ghulam Hussain, and Xiawa Wang. (2025). “Analyzing temperature-induced phase transitions in $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$.” *17th International Conference on Materials Chemistry*.

mheatmap | [\[GitHub\]](https://github.com/juntangwang/mheatmap) | **680+** Stars | Sole developer: a Python pkg. for proportional heatmap visualization and spectral reordering.

RESEARCH EXPERIENCE

ML Research: Graph Methods, Generative Models, and Biomedical AI

Mar 2025 – Present

Signature Work; Research Assistant; PI: Prof. Shixin Xu, Prof. Dongmian Zou

Kunshan, China

- Produced 10 first/co-first-author papers across biomedical AI, graph ML, neural rendering, and network security.
- Developed MixConfig (configuration-based feature augmentation) and PPIDM-II (physics-informed diffusion with abstention); proposed Variability-Gated Encoder improving joint accuracy 4.7% over Transformer on 65K-strain 16S rRNA benchmark.

EEG/EMG Time-Series Modeling for Neuroscience

Mar 2025 – Aug 2025

Summer Research Scholar; PI: Prof. Shu Kit Eric Tam

Kunshan, China

- Built a CNN achieving 90%+ accuracy on mouse EEG/EMG vigilance-state data, benchmarked against 10+ methods.
- Co-authored one SCI-journal manuscript (under review) and presented findings at a national neuroscience conference.

Graph and Spectral Methods for Hyperspectral Image Analysis

Jun 2024 – Dec 2024

Summer Research; Independent Study; PI: Prof. Xiaobai Sun

Durham, NC

- Benchmarked 5+ community-detection and clustering algorithms across 10+ hyperspectral imaging datasets.
- Released mheatmap and sgtsnepi, a pure-Python sklearn-compatible SG-t-SNE graph embedding implementation.

WORK EXPERIENCE

Quantitative Researcher, Intern

May 2026 – Jul 2026

Dahan Intelligence (~\$1B AUM self-managed quant fund)

Beijing, China

Machine Learning Engineer, Intern

Mar 2026 – May 2026

Siemens Healthineers

Shanghai, China

- Applied an InceptionTime 1-D CNN for CT X-ray tube fault diagnosis from kV/mA signals; 97% accuracy on 5 fault classes.
- Built a BERT-NER + Bedrock LLM pipeline on 200+ service docs: 99.4% PII recall, 89.4% CT-component accuracy.

ADDITIONAL INFORMATION

Service: Reviewer, NeurIPS 2026; Resident Assistant at DKU, Orientation Peer at Duke, Kendo Club Training Leader

Honors: Dean’s List with Distinction (Fall ’23, Spring ’24, Fall ’24); Kaggle Bronze (Stanford RNA 3D Folding)

Skills: Python, SQL, MATLAB, Julia, Java, C; scikit-learn, PyTorch, NumPy, pandas; PostgreSQL, MongoDB; Git, $\LaTeX$

Teaching: TA for CS 521: Matrix/Graph/Network Analysis (**Graduate**), MATH 302 Numerical Analysis, MATH 101 Calculus